TOPIC NAME : _____

DAY : _____

TIME : DATE : / /

QUANTUM MECHANICAL APPROACH OF ATOMIC STRUCTURE

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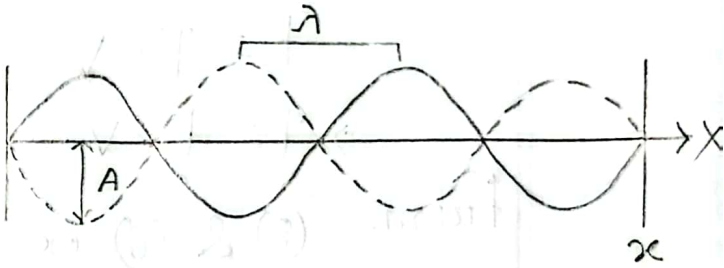
TOPIC NAME : _____ DAY : Faria - 50

TIME : DATE : / /

□ What is wave function ? Deduce the general expression for Schrodinger wave equation for a standing wave motion of a particle with mass m defining each term in it.

Ans:

A wave function refers to a mathematical description of a particle's quantum state as a function of spin, time, momentum and position. ψ is the wave function symbol.



For a standing wave, in one dimension, the wave equ. is given by,

$$\psi = A \sin 2\pi \frac{x}{\lambda} \quad \text{-----(1)}$$

where,

ψ = Value of the quantity whose magnitude is varying in wave manner.

λ = wave length

A = maximum amplitude

x = space - co-ordinate

$$\frac{d\psi}{dx} = A \cos 2\pi \frac{x}{\lambda} \left(\frac{2\pi}{\lambda}\right)$$

$$\Rightarrow \frac{d^2\psi}{dx^2} = -A \sin 2\pi \frac{x}{\lambda} \left(\frac{2\pi}{\lambda}\right)^2$$

$$\Rightarrow \frac{d^2\psi}{dx^2} = -\frac{4\pi^2}{\lambda^2} \cdot \psi$$

$$\Rightarrow \frac{d^2\psi}{dx^2} + \frac{4\pi^2}{\lambda^2} \cdot \psi = 0 \quad \text{-----(2)}$$

According to De. Broglie for a particle,

$$\lambda = \frac{h}{mv}$$

h = Plank's constant

m = mass of particle

v = velocity

$$\Rightarrow v = \frac{h}{m\lambda} \quad \text{-----(3)}$$

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Again, Kinetic energy, $T = \frac{1}{2} mv^2$

$$= \frac{1}{2} m \cdot \frac{h^2}{m^2 \lambda^2}$$

$$T \Rightarrow \frac{1}{2} \frac{h^2}{m\lambda^2} \text{ ----(4)}$$

$$\therefore \frac{1}{\lambda^2} = \frac{2m}{h^2} \cdot T \text{ ----(5)}$$

Total energy of a system,

$E = T + V$ $\quad$ V = Potential energy

$\Rightarrow T = E - V$ ----(6)

From, (5) & (6) we have,

$$\frac{1}{\lambda^2} = \frac{2m}{h^2} (E - V) \text{ ----(7)}$$

Putting the value of (7) into (2),

$$\frac{d^2\psi}{dx^2} + \frac{4\pi^2 \cdot 2m}{h^2} (E - V) \cdot \psi = 0$$

$$\Rightarrow \frac{d^2\psi}{dx^2} + \frac{8\pi^2 m}{h^2} (E - V) \psi = 0$$

For, three dimensional space, the wave equation of the particle would be,

$$\frac{d^2\psi}{dx^2} + \frac{d^2\psi}{dy^2} + \frac{d^2\psi}{dz^2} + \frac{8\pi^2 m}{h^2} (E - V) \psi = 0$$

$$\Rightarrow \nabla^2 \psi + \frac{8\pi^2 m}{h^2} (E - V) \psi = 0$$

where, $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$

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TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

□ What is Radial & Angular wave functions? Provide a graphical comparison between Radial wave function (R) and Radial probability function (r^2R^2) for 1s orbital of H.

Ans:

The wave function that describes the electron's probability distribution as a function of distance from nucleus (r) & determines the size of orbital is called radial wave function. It depends on the principal quantum number (n) and the azimuthal quantum number (l).

$g(r) = \frac{p(r)}{p_0}$; $p(r)$ = Local particles density at a distance (r) from a reference particle

p_0 = Average " " of the system.

The wave function that describes the electron's probability distribution as a function of angles (θ & ϕ) and determines the shape of the orbital is known as Angular wave function. It depends on the azimuthal quantum number (l) and the magnetic quantum number (m).

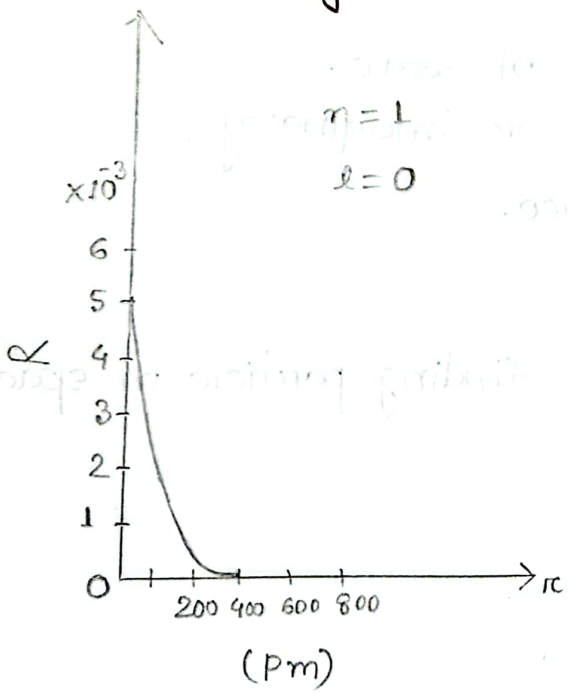


Fig: Radial wave function (R) for 1s orbital of H.

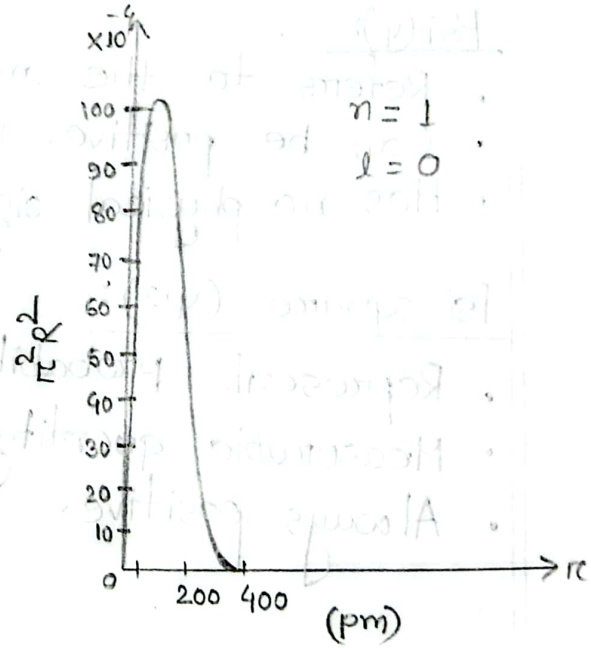


Fig: Radial probability function (r^2R^2) for 1s orbital of H.

TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

From the graph it is shown that,
The Radial wave function $R(r)$ decreases exponentially as distance from the nucleus (r) d increases, whereas the Radial Probability function (r^2R^2) starts from zero at the nucleus, reaches a maximum at a certain distance and then decreases to zero at larger distances.

□ Why the wave function ψ is squared? Point out the significance of ψ and ψ^2.

Ans:

The wave function (ψ) is squared to find out the probability of finding the particle in a specific region of space ensuring the value to be always positive as probability can not be negative.

Psi (ψ):

- Refers to the amplitude of wave.
- Can be positive, negative or imaginary.
- Has no physical significance.

Psi square ($\psi 2$):

- Represents probability of finding particle in space.
- Measurable quantity
- Always positive.

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TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

□ Define Angular function $\Theta(\theta)\Phi(\phi)$ & Radial function $R(r)$. What part of an orbital is signified by these two factors.

Ans: [* Definition has been discussed *]

These functions together define the complete shape and size of an orbital.

The angular function $\Theta(\theta)\Phi(\phi)$ gives the orbital its characteristic shape (s, p, d, f etc).

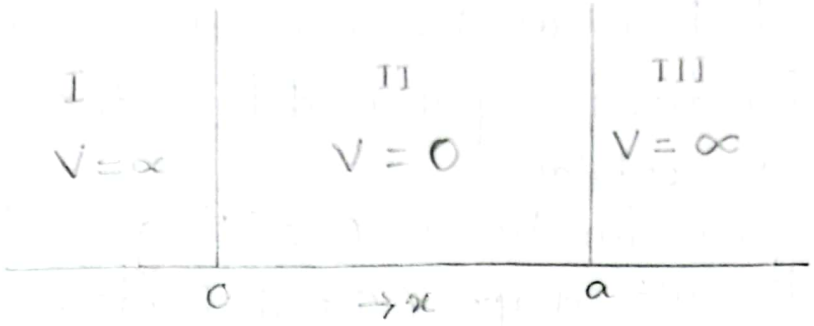
The radial function determines the orbital's size and how the electron density spreads out from the nucleus.

Thus, by combining these two functions, we can represent accurately the three-dimensional probability distribution of an electron within an atom.

□ Show that if $\psi = A\sin\pi x + B\cos sx$ (where A, B, π and s are constants), then the wave equation for a particle in a 1-dimensional box is $\psi = A\sin\pi x$.

Ans:

[P.T.O]



To be the particle within the box we have to impose the boundary conditions :-

The potential out of the box i.e. at region I & III is infinity, for $x<0$ & $x>a$; $V=\infty$

And potential at region II is zero

i.e. $0<x<a$, $V=0$.

Given, $\psi = A\sin\pi x + B\cos Sx$ ------ (1)

where, A & B, π & S are constants.

For the boundary conditions it is evident that $\psi=0$ at $x=0$ and also at $x=a$.

Using the boundary conditions in equ (1).

when $x=0$,

$$0 = A\sin\pi.0 + B\cos S.0$$

$$\Rightarrow B\cos 0^\circ = 0$$

$$\Rightarrow B\times 1 = 0$$

$$\Rightarrow B = 0$$

$$\therefore \psi = A\sin\pi x$$

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TOPIC NAME : _____ DAY : Farzia - 50

TIME : DATE : / /

□ State the Zero-point energy.

Ans:

Zero-point energy refers to the lowest possibile energy that a quantum mechanical system may have, which is the energy of it's ground state.

The energy of a particle in one dimensional box is

$$E = \frac{n^2h^2}{8a^2m}$$

The minimum energy possible when $n = 1$,

Thus the zero-point energy of that particle, $E = \frac{h^2}{8a^2m}$.

For, three dimension the zero point energy of the system would be, $E = \frac{h^2}{8a^2m}(n_x^2 + n_y^2 + n_z^2)$; As at zero point energy, $n_x = n_y = n_z = 1$

$$= \frac{h^2}{8a^2m}(1+1+1)$$

$$= \frac{3h^2}{8a^2m}$$

□ What is the physical interpretation of the wave function? Mention the acceptable conditions of a wave function ψ.

Ans:

The wave function is a fundamental concept in quantum mechanics that provides a complete description of a quantum system. It is denoted by psi (ψ). The wave function itself a complex-valued function and it's square gives the probability density of finding a particle in a particular state or position. For example, $\psi(x)$ describes the state

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of a particle & $|\psi(x)|^2$ gives the probability density of finding the particle at position x.

The acceptable conditions of a wave function are:

- ψ must be single valued. This arrises from the fact that the probability of finding a particle is proportional to ψ^2 and if it is not single valued there would be two or more distinct probabilities of finding the particle at the same point in space which is obviously impossible.
- ψ must be finite. The value ψ^2 must be finite since it represents probability.
- ψ must be continuous.

Therefore, ψ must fullfill all the conditions of a well behaved function.

□ How many radial nodes does each of the following orbital possess : (i) 2s (ii) 4s (iii) 3p (iv) 5d (v) 1s (vi) 4f

Ans:

The number of radial nodes for a given orbital is given by: $(n - l \mp 1)$.

(i) For 2s,

$n = 2$, $l = 0$

$\therefore$ Radial node $= 2 - 0 \mp 1 = 1$

TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

(ii) For 4s,

$n=4 \quad l=0$

$\therefore$ radial node $= 4-0-1 = 3$

(iii) For, 3p,

$n=3 \quad l=1$

$\therefore$ radial node $= 3-1-1 = 1$

(iv) For 5d,

$n=5, \quad d=2$

$\therefore$ radial node $= 5-2-1 = 2.$

(v) For 1s,

$n=1 \quad l=0$

$\therefore$ radial node $= 1-0-1 = 0$

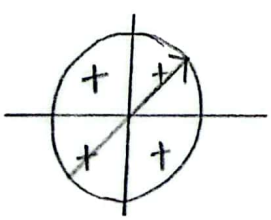
(vi) For 4p,

$n=4, \quad l=1$

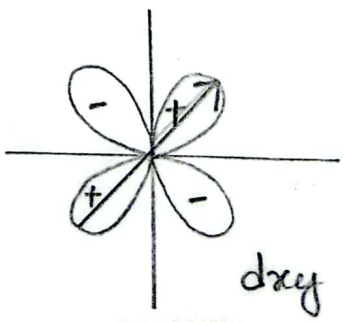
$\therefore$ radial node $= 4-1-1 = 2$

□ What is meant by gerade and ungerade orbitals ?

If on moving from any point A to the equivalent point B on the opposite side of the center, the sign of the wave function does not change, the orbital is said to be gerade. Ex : s and d orbitals.



s



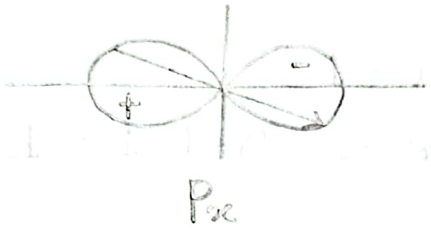
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TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

And, if the sign of the wave function change, the orbital is called ungerade. p and f orbitals are ungerade as they are antisymmetric with respect to inversion and the sign changes on going from A to B.



P_x

□ What is meant by well behaved function and normalization of wave function.

Ans:

A function is said to be well behaved function which is finite, single valued and continuous.

- Finite : The function's values must be finite everywhere in space.
- Single-valued : There must be only one value of the function at any given point in space.
- Continuous : The function must be continuous everywhere in space, except possibly at points where the potential energy is finite.

Ψ is considered as a well behaved function.

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TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

Normalization of a wave function is a concept that ensures the total probability of finding a particle in all possible states or positions equals one. For a wave functio to be normalized, the integral of the probability density over all space must equal to one.

If a wave function/equation has two solution: ψ, $A\psi$, the

$A^2 \int \psi^2 \, dx.dy.dz = 1$ represents a normalized wave function & A is the normalizing factor.

where, $\psi \neq 1$

□ What is the significance of Schrodinger equation ?

Ans:

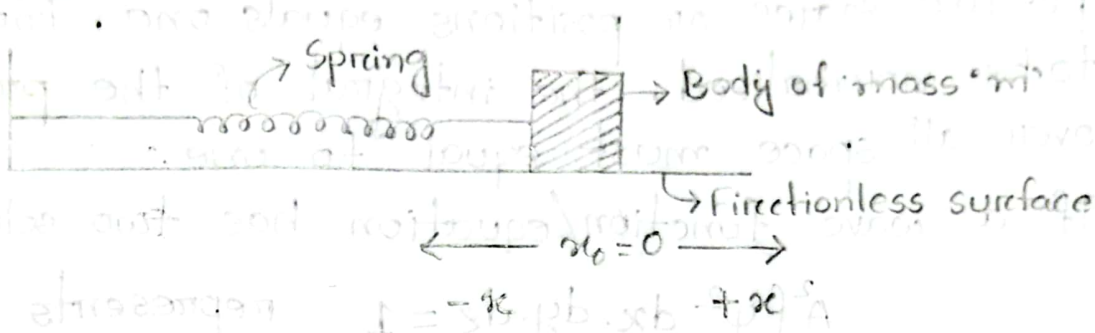
It describes the wave-particle duality and contains all possible information about quantum system. Solving the equation properties like probability of finding a particle in a certain location, its energy, momentum etc can be determined.

TOPIC NAME : _____ DAY : Farria - 50

TIME : DATE : / /

□ Schrodinger Equation for different conditions:

□ For Simple hermonic oscillator:



Let us consider, a simple hermonic oscillator made of a weightless spring and a body of mass 'm' attached to it and moving on a frictionless surface.

Let $x_0 = 0$ be the position of the body at normal stage. If the body is streched or extended, according to Hook's law the force is,

$$F = -Kx \quad \text{where } K = \text{Force constant}$$

Hence, the potential energy of the system is

$$V = \frac{1}{2} Kx^2$$

Putting the value of V in the general equation,

$$\frac{d^2\psi}{dx^2} + \frac{8\pi^2 m}{h^2}(E - V)\psi = 0 \quad \text{we get,}$$

$$\frac{d^2\psi}{dx^2} + \frac{8\pi^2 m}{h^2}\left(E - \frac{1}{2}Kx^2\right)\psi = 0$$

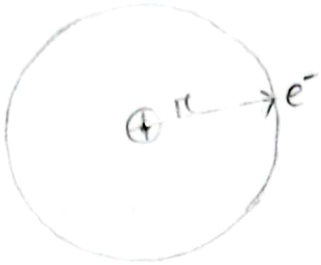
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TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

□ Hydrogen atom :

Q Write the Schrodinger wave equation for the H atom in Cartesian co-ordinate system and explain the terms in it.



The potential energy of the electron of hydrogen atom depends on r; the distance between the nucleus and the electron. To bring an electron from infinity to a distance r, the work done will be,

$$\int_{\infty}^{r} F(r)\,dr = \int_{\infty}^{r} \frac{e^2}{r^2}\,dr = -\frac{e^2}{r}$$

, which will also b the potential energ of the system.

∴ Potential energy, P.E , $V = -\frac{e^2}{r}$

Thus, Schrodinger Equation for H atom be,

$$\nabla^2\psi + \frac{8\pi^2 m}{h^2}\left(E + \frac{e^2}{r}\right)\psi = 0$$

N.B: For Hydrogen like systems (He^+, Li^{2+} etc) equ will b

$$\nabla^2\psi + \frac{8\pi^2 m}{h^2}\left(E + \frac{Ze^2}{r}\right)\psi = 0$$

where, ∇ = Laplacian operator $= \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$

ψ = wave function

h = Plank's constant

m = mass of particle

E = Total energy = (T+V)

Z = Atomic no. of atom

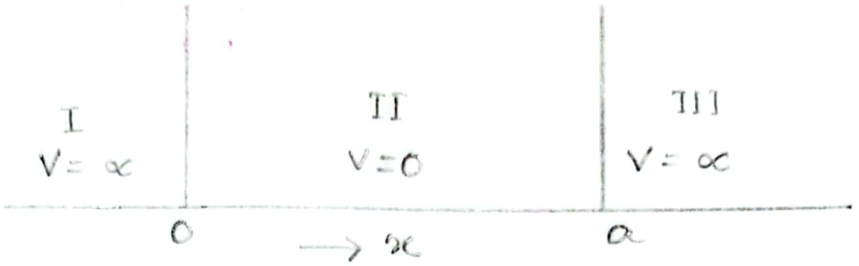
r = Distance between e^- and nucleus

CamScanner

TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

□ Particle in a box (One dimensional system) :-



To be the particle within the box, the boundary conditions to be imposed.

The potential at region I & III is infinity.

i.e. for $x<0$ & $x>a$, $V=\infty$

And the potential at region II is zero.

i.e. $0<x<a$, $V=0$

The general wave equation for one dimensional case

$$\frac{d^2\psi}{dx^2} + \frac{8\pi^2 m}{h^2}(E-V)\psi = 0 \text{ -----(1)}$$

when particle is within the box, $V=0$, thus,

$$\frac{d^2\psi}{dx^2} + \frac{8\pi^2 m}{h^2}E\psi = 0 \text{ ----(2)}$$

For particular value of E, $\frac{8\pi^2 mE}{h^2}$ is constant.

Let, $\frac{8\pi^2 mE}{h^2} = K^2$ ----(3)

Equation (2) becomes,

$$\frac{d^2\psi}{dx^2} + K^2\psi = 0 \text{ -----(4)}$$

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Equ. (4) is a standard 2nd order differential equation. And its general solution will be.

$$\psi = A\sin Kx + B\cos Kx \quad \text{------(5)}$$

where A & B are constant.

From the boundary conditions it is evident that.

$\psi = 0$ at $x = 0$ and also at $x = a$

when $x = 0$ (5) becomes,

$$0 = A\sin K.0 + B\cos.K.0$$

$$\Rightarrow B\cos 0^\circ = 0$$

$$\Rightarrow B \times 1 = 0$$

$$\therefore \psi = A\sin Kx \quad \text{-----(6)}$$

when $x = a \neq 0$ from eq. (6) we get.

$$0 = A\sin Ka$$

$$\therefore \sin Ka = 0 \quad , A \neq 0$$

$$\Rightarrow \sin Ka = \sin n\pi$$

$$\Rightarrow Ka = n\pi$$

$$\Rightarrow K = \frac{n\pi}{a} \qquad n = 1, 2, 3\text{---} \text{ but } n \neq 0$$

The equ (6) becomes,

$$\psi = A\sin\frac{n\pi x}{a} \quad \text{------(7)}$$

Again from the boundary condition it is evident that the probability of finding the particle within the box is unity. So,

$$\int_0^a \psi^2 dx = 1$$

$$\Rightarrow \int_0^a A^2 \sin^2\frac{n\pi x}{a} dx = 1$$

$$\Rightarrow \frac{1}{A^2} = \int_0^a \sin^2\frac{n\pi x}{a} dx$$

CamScanner

TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

$$= \int_0^a \frac{1}{2}\left(1-\cos\frac{2n\pi x}{a}\right)dx$$

$$= \int_0^a \frac{1}{2}\,dx - \int_0^a \cos 2n\frac{\pi}{a}\,dx$$

$$\frac{1}{A^2} = \frac{a}{2} - 0$$

$$\Rightarrow A^2 = \frac{2}{a}$$

$$\Rightarrow A = \sqrt{\frac{2}{a}}$$

Value of A

Thus, $\psi = \sqrt{\frac{2}{a}} \sin n\pi\frac{x}{a}$

Using $K = \frac{n\pi}{a}$ in eq (3),

$$\frac{8\pi^2 mE}{h^2} = \frac{n^2\pi^2}{a^2}$$

$$\therefore E = \frac{n^2h^2}{8a^2m}$$ [Energy of the particle]

Graphical representation:

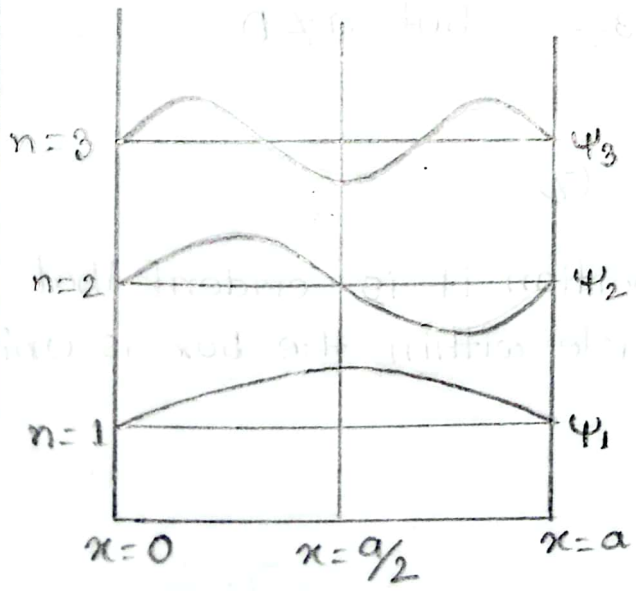


Fig: ψ

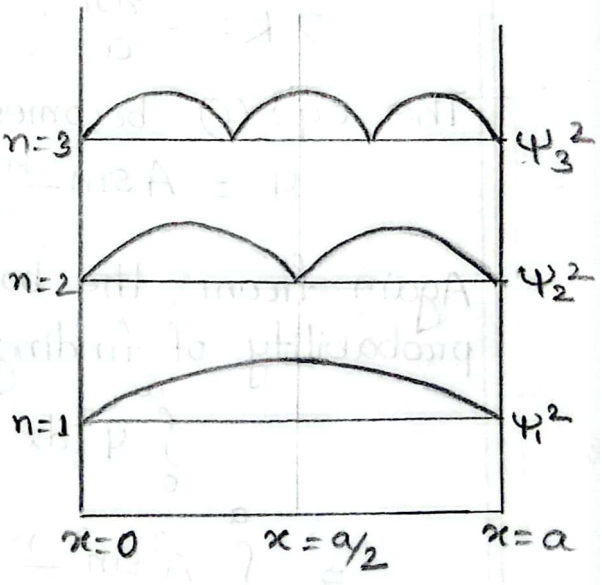


Fig: ψ²

TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

Particle in box (Three dimensional system) :

Let us consider a three dimensional box. Let the length of the box along x, y and z axes a_x, a_y and a_z respectively. Let a particle of mass 'm' and velocity 'v' is within the box which can not escape the box i.e. Potential of the particle within the box $V = 0$ and outside the box $V = \infty$.

The general Schrodinger equ. for three dimension is,

$$\frac{\partial^2\psi}{dx^2} + \frac{\partial^2\psi}{dy^2} + \frac{\partial^2\psi}{\partial z^2} + \frac{8\pi^2 m}{h^2}(E-V)\psi = 0$$

As, $V = 0$,

$$\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2} + \frac{8\pi^2 m}{h^2}E\psi = 0 \quad \text{------ (1)}$$

As the wave function ψ is dependent on the x, y, z co-ordinates,

$$\psi = \psi_x \cdot \psi_y \cdot \psi_z \quad \text{------ (2)}$$

Differentiating eq. (2) w.r.t to x,

$$\frac{d}{dx}\psi = \psi_y \cdot \psi_z \cdot \frac{d}{dx}\psi_x$$

$$\Rightarrow \frac{d^2}{dx^2}\psi = \psi_y \cdot \psi_z \cdot \frac{d^2}{dx^2} \cdot \psi_x \quad \text{---- (3)}$$

Similarly,

$$\frac{d^2}{dy^2}\psi = \psi_x \cdot \psi_z \cdot \frac{d^2}{dy^2}\psi_y \quad \text{---- (4)}$$

$$\frac{d2}{dz^2}\psi = \psi_x \cdot \psi_y \cdot \frac{d^2}{dy^2} \cdot \psi_z \quad \text{---- (5)}$$

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Using (3), (4) & (5) we get,

$$\psi_y \cdot \psi_z \frac{d^2}{dx^2} \cdot \psi_x + \psi_x \cdot \psi_z \cdot \frac{d^2}{dy^2} \cdot \psi_y + \psi_x \cdot \psi_y \cdot \frac{d^2}{dz^2} \cdot \psi_z + 8\frac{\pi^2 m}{h^2} \cdot E \psi_x \psi_y \psi_z = 0$$

$$\Rightarrow \frac{1}{\psi_x} \cdot \frac{d^2}{dx^2} \psi_x + \frac{1}{\psi_y} \cdot \frac{d^2}{dy^2} \cdot \psi_y + \frac{1}{\psi_z} \cdot \frac{d^2}{dz^2} \cdot \psi_z + \frac{8\pi^2 m}{h^2} E = 0$$

[Dividing by $\psi_x \psi_y \psi_z$]

$$\Rightarrow \frac{1}{\psi_x} \cdot \frac{d^2}{dx^2} \cdot \psi_x + \frac{1}{\psi_y} \cdot \frac{d^2}{dy^2} \cdot \psi_y + \frac{1}{\psi_z} \cdot \frac{d^2}{dz^2} \cdot \psi_z = -\frac{8\pi^2 m E}{h^2}$$

$$\Rightarrow \frac{h^2}{8\pi^2 m} \left(\frac{1}{\psi_x} \cdot \frac{d^2}{dx^2} \psi_x + \frac{1}{\psi_y} \cdot \frac{d^2}{dy^2} \cdot \psi_y + \frac{1}{\psi_z} \cdot \frac{d^2}{dz^2} \cdot \psi_z \right) = -E$$

The energy E can be resolved into three components,

$$E = E_x + E_y + E_z$$

where,

$E_x \propto \frac{1}{2} m v_x^2$
$E_y \propto \frac{1}{2} m v_y^2$
$E_z \propto \frac{1}{2} m v_z^2$

v_x, v_y & v_z are the velocities of the particle along x, y & z axes.

So, $$\frac{h^2}{8\pi^2 m} \cdot \frac{1}{\psi_x} \cdot \frac{d^2}{dx^2} \psi_x = -E_x$$

$$\Rightarrow \frac{1}{\psi_x} \cdot \frac{\partial^2}{\partial x^2} \psi_x + \frac{8\pi^2 m}{h^2} \cdot E_x = 0$$

$$\Rightarrow \frac{\partial^2}{\partial x^2} \psi_x + \frac{8\pi^2 m}{h^2} \cdot E_x \cdot \psi_x = 0$$

CamScanner

TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

Similarly,

$$\frac{\partial^2}{\partial y^2}\psi_y + \frac{8\pi^2 m}{h^2} \cdot E_y \cdot \psi_y = 0$$

$$\frac{\partial^2}{\partial z^2} \cdot \psi_z + \frac{8\pi^2 m}{h^2} \cdot E_z \cdot \psi_z = 0$$

The normalized solution of these three equations are,

$$\psi_x = \sqrt{\frac{2}{a_x}} \cdot \sin \frac{n_x \pi x}{a_x}, \quad \psi_y = \sqrt{\frac{2}{a_y}} \cdot \sin \frac{n_y \pi y}{a_y}$$

$$\psi_z = \sqrt{\frac{2}{a_z}} \sin \frac{n_z \pi z}{a_z}$$

And associated energy,

$$E_x = \frac{n_x^2 h^2}{8 m a_x^2} \quad E_y = \frac{n_x^2 h^2}{8 m a_y^2}, \quad E_z = \frac{n_z^2 h^2}{8 m a_z^2}$$

$$\therefore \psi = \psi_x \cdot \psi_y \cdot \psi_z$$

$$= \sqrt{\frac{2}{a_x}} \sin \frac{n_x \pi x}{a_x} \cdot \sqrt{\frac{2}{a_y}} \sin \frac{n_y \pi y}{a_y} \cdot \sqrt{\frac{2}{a_z}} \sin \frac{n_z \pi z}{a_z}$$

$$\psi = \sqrt{\frac{8}{a_x a_y . a_z}} \cdot \sin\left(\frac{n_x \pi x}{a_x}\right) \cdot \sin\left(\frac{n_y \pi y}{a_y}\right) \cdot \sin\left(\frac{n_z \pi z}{a_z}\right)$$

$$E = E_x + E_y + E_z = \frac{n_x^2 h^2}{8 m a_x^2} + \frac{n_x^2 h^2}{8 m a_y^2} + \frac{n_z^2 h^2}{8 m a_z^2}$$

If $a_x = a_x = a_y = a$

- $\psi = \sqrt{\frac{8}{a^3}} \sin\left(\frac{n_x \pi x}{a_x}\right) \cdot \sin\left(\frac{n_y \pi y}{a_y}\right) \cdot \sin\left(\frac{n_z \cdot \pi z}{a_z}\right)$
- $E = \frac{h^2}{8 m a^2}\left(n_x^2 + n_y^2 + n_z^2\right)$

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TIME : DATE : / /

□ Use Slater's rule to estimate values of Zeff for (i) a 4s (ii) a 3d electron in a V atom. (Z = 23).

Ans:

Effective nuclear charge:

The effective nuclear charge is the net positive charge experienced by an electron towards the nucleus in an atom. It is denoted by Zeff

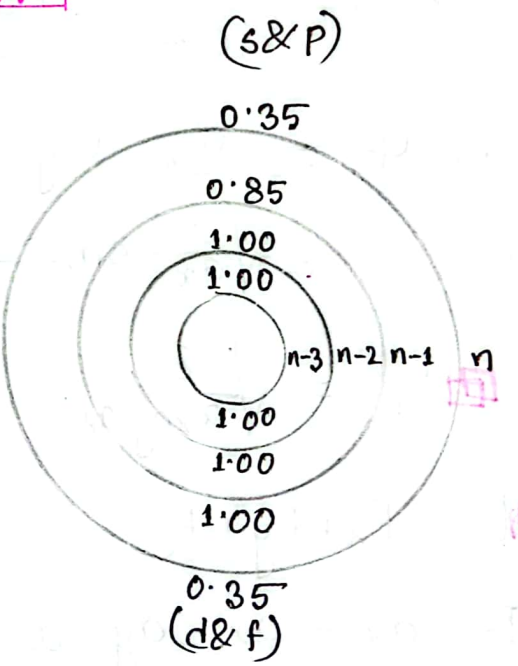
$$Z_{eff} = Z - \sigma/s$$

where, Z = nuclear charge

σ/s = shielded or screening constant.

Tricks for Slater's rule !!

e⁻ in orbital	Per electron in orbital		
	n	n−1	n−2, n−3
s / p	0·35	0·85	1·00
d/f	0·35	1·00	1·00



TOPIC NAME : _____

DAY : _____

TIME : DATE : / /

For V (23) → $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^3\ 4s^2$







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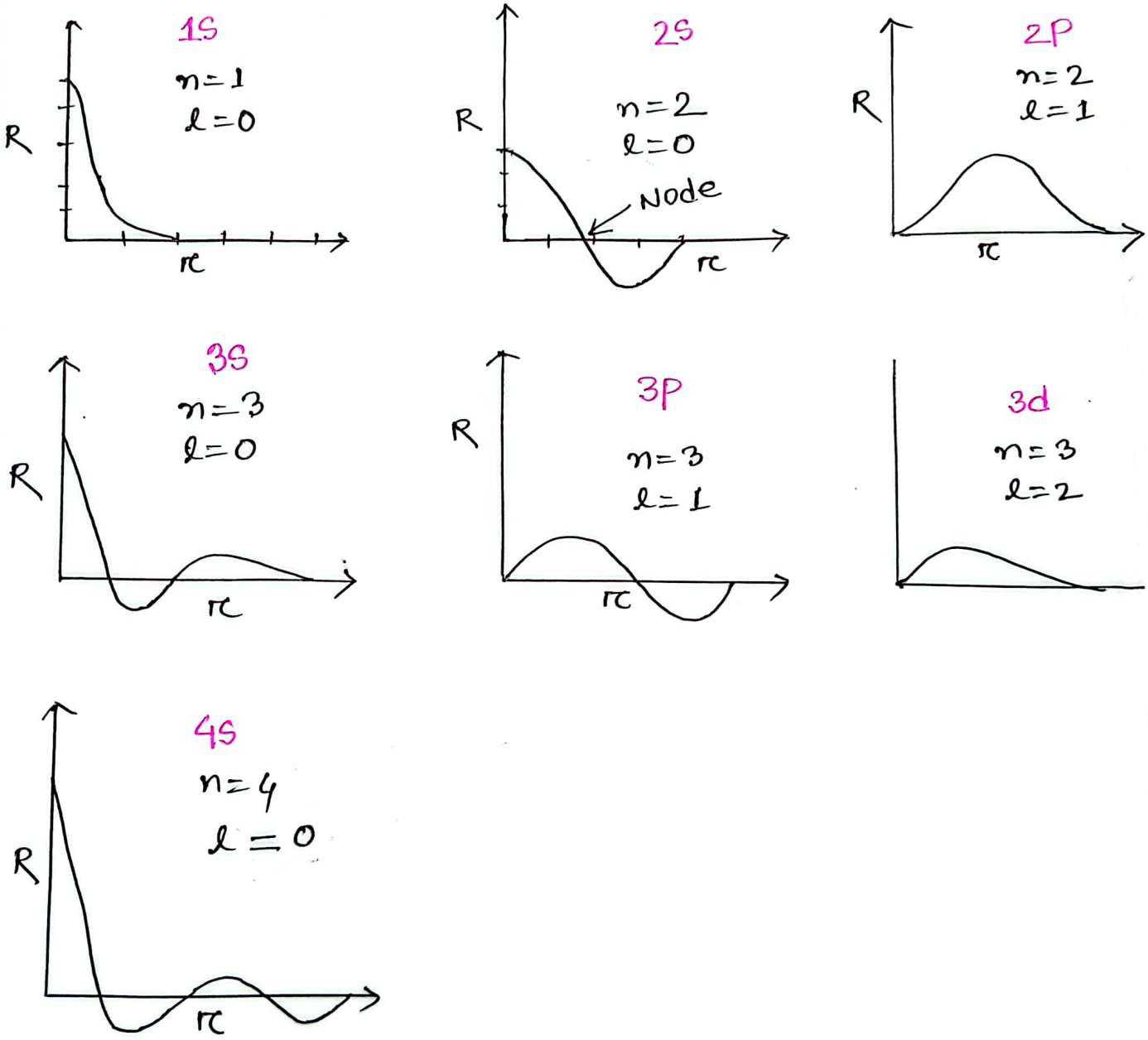
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TIME : DATE : / /

Q Summarize and represent in figures (i) radial part of a wave function (ii) radial probability distribution curves of the electron wave function for different electronic states of the hydrogen atoms.

Ans:

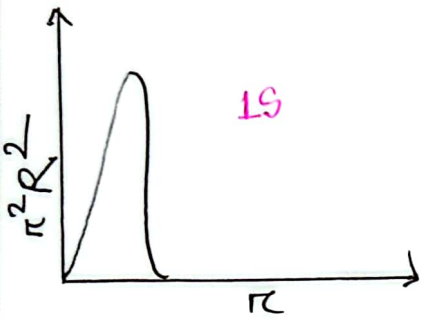
Distribution curve for radial wave function:

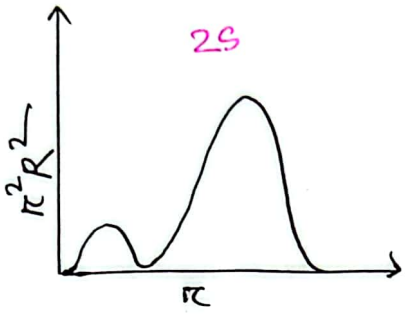


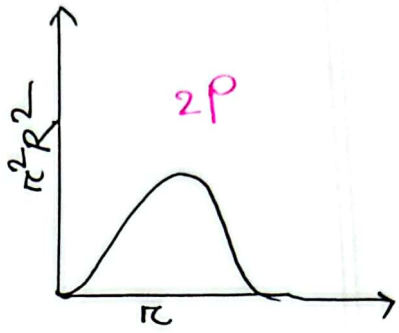
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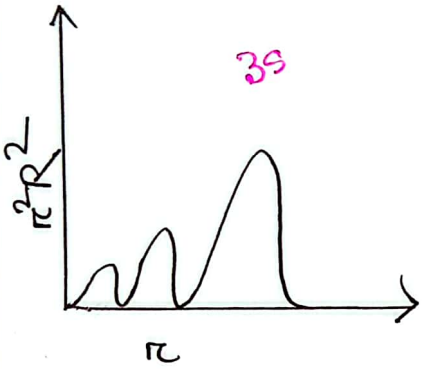
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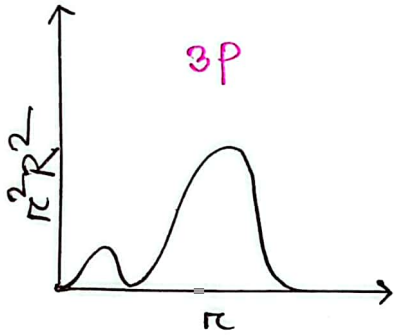
Radial probability distribution curves :

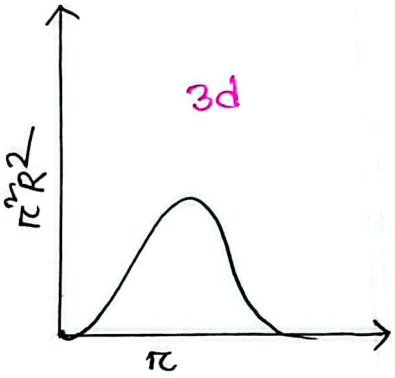


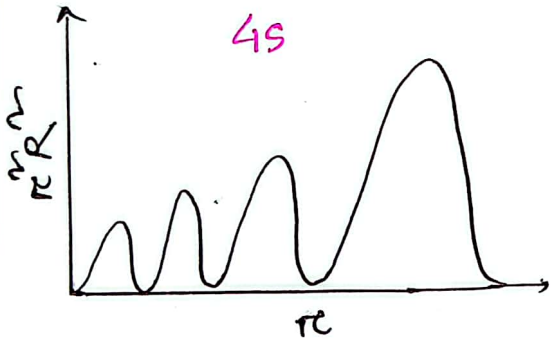












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